

Reading and Writing Files in Python

File handling is a fundamental skill. Python makes this easy with the built-in `open()` function and the `with` statement.

1. The `with` Statement (Context Managers)

In older Python code, you might see files opened and closed manually. However, if an error occurs before `.close()` is reached, the file stays open in memory.

The **`with` statement** is a **Context Manager**. It automatically closes the file for you as soon as the code block finishes, even if an error occurs.

2. Writing to a File

Common modes for `open()` :

- `'w'` : **Write** (Overwrites the file if it exists, or creates a new one).
- `'a'` : **Append** (Adds text to the end of the existing file).
- `'r'` : **Read** (Default mode).

```
In [... # Writing to a file
with open("example.txt", "w") as file:
    file.write("Hello World!\nThis file was created u
# file.write("")
```

```
with open("example.txt", 'a') as file:
    file.write('\nThis text was appended later.')
# print("File written successfully!")
```

```
Out[... 73
```

```
Out[... 30
```

```
In [... # csv files are plain text files that stores values s
# this is a widely used format and it's especially us
```

```
# generally, I would advise to use pandas for handling
with open('table.csv', 'w') as file:
    file.write('Number,Square\n')
    for i in range(1,11):
        file.write(f'{i},{i**2}\n')
```

Out[... 14

Out[... 4

Out[... 4

Out[... 4

Out[... 5

Out[... 5

Out[... 5

Out[... 5

Out[... 5

Out[... 5

Out[... 7

3. Reading from a File

You can read the entire content at once, or iterate line by line (which is better for large files).

```
In [... # # Reading the entire file
# with open("table.csv", "r") as file:
#     content = file.read()
#     # print(f"Full Content of {file.name}:")
#     print(content)

# Reading line by line (Memory efficient)
with open("table.csv", "r") as file:
    print(f"Line by Line content of {file.name}:")
    i = 0
    for line in file:
        print(f"{line.strip()}") # .strip() removes t
```

Line by Line content of table.csv:

Number,Square

1,1

2,4

3,9

4,16

5,25

6,36

7,49

8,64

9,81

10,100

In [...]

```
foo = list('abcd')
for i,letter in enumerate('alsdfjld'): # string is a
    print(i,letter)
```

0 a

1 l

2 s

3 d

4 f

5 j

6 l

7 k

8 d

Summary of file modes

Mode	Description	File Pointer Position
'r'	Read (default)	Start of file
'w'	Write (overwrites)	Start of file
'a'	Append (adds to end)	End of file
'r+'	Read and Write	Start of file

In [...]

```
with open('num.txt', 'w') as file:
    for i in range(5):
        file.write(f'{i}\n')
```

Out[... 2

Out[... 2

Out[... 2

Out[... 2

Out[... 2